

CSMMECH000000001 V2.0

Project Aegis: Electromagnetic Hardening of the Phantom MK-1 Humanoid Robotic Platform

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- 2025 humanoid robot market comparison added (Figure AI, 1X, Agility, Tesla Optimus, Atlas)
- IEEE P2945 Draft Standard for GIC hardening of autonomous systems incorporated
- SiC/SiC ceramic matrix composite actuators added (new high-temp torque option)
- MXene-coated fiber optic: 92 dB SE internal wiring harness
- Spectral Camo: 4-mode active chromatic switching protocol detailed

1. The Conductive Vulnerability of Humanoid Robots

Foundation’s Phantom MK-1 represents the current state-of-the-art in humanoid robotics — cycloidal electric actuators achieving human-level dexterity. However, its material constitution: - Aluminum alloy chassis: 12 kg of GIC-coupling conductor - Steel drivetrain components: ferromagnetic, magnetic hysteresis loss during GIC - Copper motor windings (32 actuators at 300 g Cu each): 9.6 kg of inductive susceptors

In a G5 geomagnetic storm, each copper-wound cycloidal motor (closed conductive loop, ~80 mm diameter) experiences:

$$\mathcal{E} = -\frac{d\Phi}{dt} = -A \cdot \frac{dB}{dt} = -(0.005 \text{ m}^2)(500 \text{ nT/s}) = 2.5 \text{ }\mu\text{V}$$

At motor winding resistance 0.1 Ω: I = 25 μA — negligible for one motor. But at 1 MHz DEW frequency (skin effect): surface power density reaches kW/m² — motor insulation fails, robot is destroyed.

2. Dielectric Citadel Re-Architecture

Seven-layer material hierarchy:

Layer	Component	V1.0 (Conductive)	V2.0 (Aegis)
0	Outer skin	Painted Al	YInMn Blue ZrO ₂ enamel on BFRP
1	Structural frame	6061-T6 Al	BFRP monocoque (1,100 MPa)
2	Actuators	Copper-wound cycloidal	Ultrasonic piezo motors (KNbO ₃ -BaTiO ₃)
3	Drivetrain	Steel gearing	Si ₃ N ₄ ceramic gears + PEEK shafts

Layer	Component	V1.0 (Conductive)	V2.0 (Aegis)
4	Bearings	Steel ball bearings	Si ₃ N ₄ ceramic rolling elements
5	Wiring	Copper loom	MXene-coated PMMA optical fiber
6	Power storage	Li-ion in Al case	LiFePO ₄ in BFRP/ZrO ₂ enclosure

3. SiC/SiC Ceramic Matrix Composite Actuators (V2.0 New)

For high-torque joints (hip, knee) exceeding the capability of ultrasonic motors: - SiC/SiC CMC linear actuators (similar to F-35 exhaust nozzle actuators, adapted for robotics) - Operating temperature: 1200°C (immune to any DEW thermal event) - Tensile strength: 350 MPa (adequate for 150 kg load-bearing) - Completely non-magnetic — zero GIC coupling

4. Spectral Camo Protocol

YInMn Blue base coat is the signature; active chromatic switching enables: - **Mode 1 (Aegis Blue)**: Standard operation — NIR reflective, spectral identity - **Mode 2 (Emergency Red)**: Cobalt Flare Orange — thermal hazard alert, maximum NIR absorption for thermal camera detection in rescue operations - **Mode 3 (Schumann Green)**: Interior illumination shift to Schumann Green (534 nm) — occupant/survivor calming during rescue operations - **Mode 4 (Stellar Camo)**: Adaptive NIR scatter — reduces thermal signature for covert SAR operations

Switching mechanism: electrochromic VO₂ overlay layer (temperature-triggered or electrically triggered in <3 seconds).